14. $O_2^{2-}$ is diamagnetic. Additional two electrons paired in antibonding molecular orbital $\pi^*_{2P_y}$ and $\pi^*_{2P_z}$ (option (b))

15. Bond order = $\frac{1}{2}(n_b - n_a)$

$2.5 = \frac{1}{2}(8 - n_a)$

$\Rightarrow 5 = 8 - n_a$

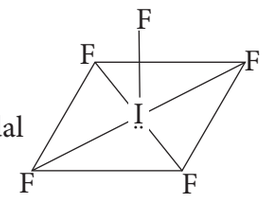
$\Rightarrow n_a = 8\text{-}5 = 3$ (option a)

16. $IF_5$ - 5 bond pair + 1 lone pair

hybridisation $Sp^3d^2$

Shape

Square pyramidal




17. Correct statement :

All five $Sp^3d$ hybrid orbitals are equivalent

$\therefore$ incorrect statement : Option (c)

18. $SeF_4$, $XeO_2F_2$ - $Sp^3d$ hybridisation,

option (a) T-shaped, one line pair on central atom.

19. $H_2O$ - Central atom $Sp^3$ hybridised

$NO_2^-$ - Central atom $Sp^2$ hybridised

$BF_3$ - Central atom $Sp^2$ hybridised

$NH_2^-$ - Central atom $Sp^3$ hybridised

option (c) is correct.

20. $NO_3^-$ - $Sp^2$ hybridisation, planar

$H_3O^+$ - $Sp^3$ hybridisation, pyramidal

option (a) is correct.

21. $\overset{Sp^3}{CH_3} - \overset{Sp^2}{CH} = \overset{Sp}{C} = \overset{Sp^3}{CH} - \overset{Sp^3}{CH_3}$ (option a)

22. $XeF_2$ is isostructural with $ICl_2^-$

23. $\overset{Sp^3}{CH_4}$, $\overset{Sp^3}{CH_3} - CH_3$, $\overset{Sp^2}{CH_2} = CH_2$, $\overset{Sp}{CH} \equiv CH$

25, 25, 33.3, 50

option (a) is correct.

24. $Co_2$ - Linear

$C_2H_2$ - Linar

option (c) is correct.

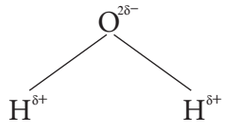
25. l.p - l.p > l.p - b.p > b.p - b.p

option (c) is correct

26. $ClF_3$ - $Sp^3d$ hybridisation, 'T' shaped

option (c) is correct.

27.




option (d)

28. Correct statement is - the resonance hybrid should have lower energy than any of the contributing structure

Hence is correct statement is option (c)

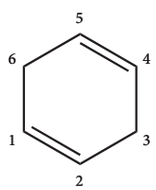
29. $NH_4Cl$ (option (a))

30. 4U

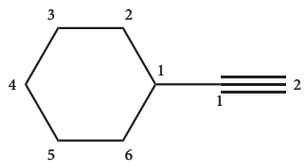
## UNIT - 11

## EVALUATE YOURSELF - KEY

**2.**

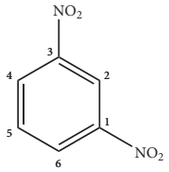



cyclohexa- 1,4-diene

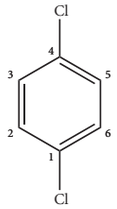



ethynyl cyclohexane

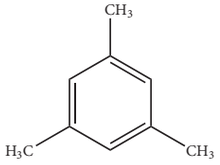
**3.**




1,3- dinitrobenzene




p-dichlorobenzene




1, 3, 5 trimethylbenzene



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